

MIRA KRISHNAIAH

Artificial Intelligence and Mechanical Engineering student at UPenn, passionate about bridging AI research and deployable hardware. Published machine learning researcher with start-up pace experience shipping mechanical systems from rapid prototyping to competition-ready hardware. Seeking roles at the intersection of embodied AI, robotics automation, and physical system design.

EDUCATION

University of Pennsylvania

May 2029 (Expected)

Candidate for B.S.E. in Artificial Intelligence (Minor: Mechanical Engineering). Relevant coursework includes:

- **Mathematics:** Multivariate Calculus, Linear Algebra, Number Theory, Discrete Probability, Models of Computation, Computable Structures, Linear Topology, Voting Theory, Information Theory, Game Theory.
- **Computer Science:** Computing Systems Design, Artificial Intelligence Lab.

PRODUCT AND RESEARCH EXPERIENCE

ModLab | University of Pennsylvania

Mechanical Engineering Researcher

September 2025 - Present

- Redesigned fatigue testing rig for Twisted String Actuators: reduced link weight, integrated magnetic encoder for autonomous cycle tracking, fabricated and assembled test setup

The Missfits (Team 6418) | FIRST Robotics

Team President and Head of Design

August 2023 - August 2025

- Rebuilt a nearly-dissolved program after COVID setbacks, securing a workshop, recruiting 10+ members, driving fundraising, and raising technical training to certification level.
- Won Engineering Excellence for design process and Creativity for custom coaxial linkage arm design.
 - Led end-to-end robot development across five subteams, under 6-week product deadlines.
 - Personally CAD-modeled ~90% of all custom subsystems in Onshape for two 150lb competitive robots.
 - Executed 250+ hours of CNC machining, prototyping, and assembly of all mechanisms.
- Collaborated with industry professionals from autonomous vehicle fields.

Summer Research Academies | UCSB

Machine Learning Researcher

Summer 2023

- Published research paper on the [Curieux Academic Journal](#) (pg 145).
- Trained an ANN in Python to accurately predict an element's Young's, Bulk, and Shear Moduli with 98% accuracy, reducing alloy screening cost for additive-manufacturing in aerospace applications.

ATLAS Institute | CU Boulder

Haptic Design Researcher

Summer 2022

- Developed "ShapeUp" prototype with 16 independently actuated tiles, creating low-cost shape-changing display. Designed servo-driven rack and pinion + 3D printed scissor-lift system.

Bloom Board | Independent

Product Developer

May 2023 - June 2025

- Created display of 3D printed interactive robotic flowers to provide students an anonymous method of communicating their emotional wellbeing to school administration.
- Won 2nd place Live Conference Presentation at Polygence Symposium.

ADDITIONAL SKILLS

- **Engineering:** Onshape, Solidworks, GD&T, CNC Machining, Prototyping, 3D Printing
- **AI/Programming Languages:** Python (Machine learning, ANNs), Java, OCaml
- **Languages:** Fluent in Mandarin